

AN ULTRA-LOW POWER BANDGAP VOLTAGE AND CURRENT  
REFERENCE

By

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## An Ultra-low Power Bandgap Voltage and Current Reference

Abstract

by

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In this thesis, we developed a new low-power bandgap voltage and current reference which consumes 0.9 $\mu$ W at room temperature and provides an 821mV/1.6 $\mu$ A reference voltage and current with 63.5/246ppm/ $^{\circ}$ C accuracy. The main highlight of our work is to greatly reduce the power consumption (<1 $\mu$ W) of the bandgap voltage reference while maintaining a relatively good accuracy which makes this bandgap reference a suitable candidate for ultra-low power devices. At the same time, we provide a reference current that occupies a very small chip area, which is not available in most voltage references. This structure also has the potential to further reduce power consumption to the nanowatt level and higher accuracy.

# 1 Introduction

## 1.1 Research Background

Voltage references can be used to generate a constant voltage that is independent of supply voltage, temperature, and process. It's the most commonly used method to create a temperature-independent voltage and can be found in almost all modern electronic products. These voltage references can be a separate part or can be integrated into other components such as ADC, DAC, or voltage regulator. For example, as shown in figure 1.1, multiple

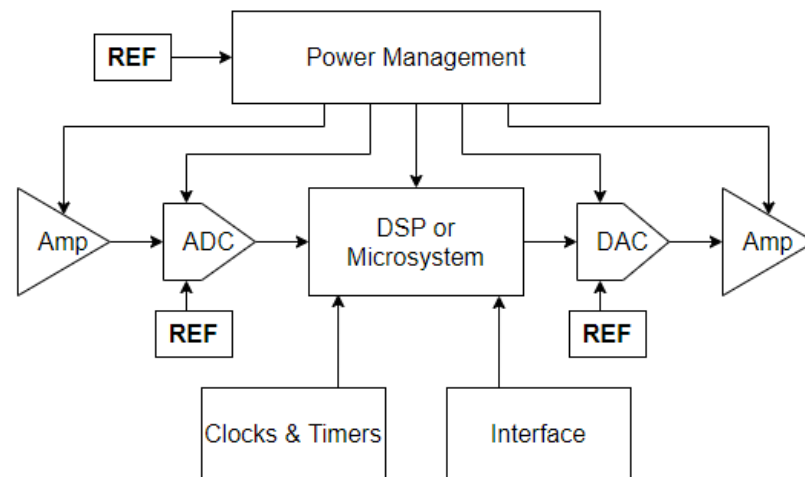


Figure 1.1 A mixed-signal system

separate voltage references may be used in a mixed-signal system to generate different voltages and avoid crosstalk through a single supply<sup>[1]</sup>. People's search for the reference voltage can be traced back to 1873, where English

engineer Josiah Latimer Clark invented Clark cell that can produce 1.457V reference voltage at 15 degrees<sup>[2]</sup>. Nowadays, there are several types of voltage references in solid-state devices and can be found in many devices including Zener-diode-based reference, bandgap reference, buried Zener reference, and XFET reference. Among them, bandgap references<sup>[3]</sup> are widely used where they reach very high accuracy and stability<sup>[4][5]</sup>.

## 1.2 Typical Application of Reference Voltage

### 1.2.1 Voltage Regulator

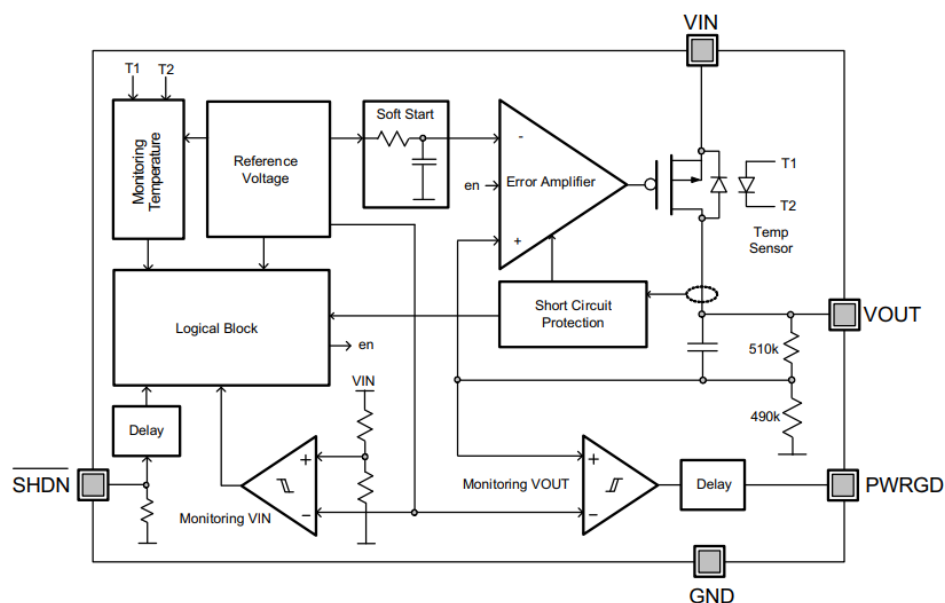


Figure 1.2 Voltage Regulator from Microchip<sup>[16]</sup>

Figure 1.2 shows voltage regulator MCP1790 from the microchip<sup>[16]</sup>. A typical voltage regulator contains three basic functional components: reference voltage, power FET, a differential amplifier (error amplifier), and some protection and monitoring components. One input of the differential amplifier

from the fraction of the output is determined by the resistor ratio of two sampling resistors. The second input to the differential amplifier is from a voltage reference. When the circuit is started, the reference voltage is first established, and the output rises with the input. When the output is close to the targeted value, the feedback voltage obtained by the sampling resistor is also close to the reference voltage. At this time, the error amplifier amplifies the small error between the output feedback voltage and the reference voltage and then amplified to the output by the power FET to form negative feedback to ensure that the output voltage is stable at the targeted value.

**functional block diagram**

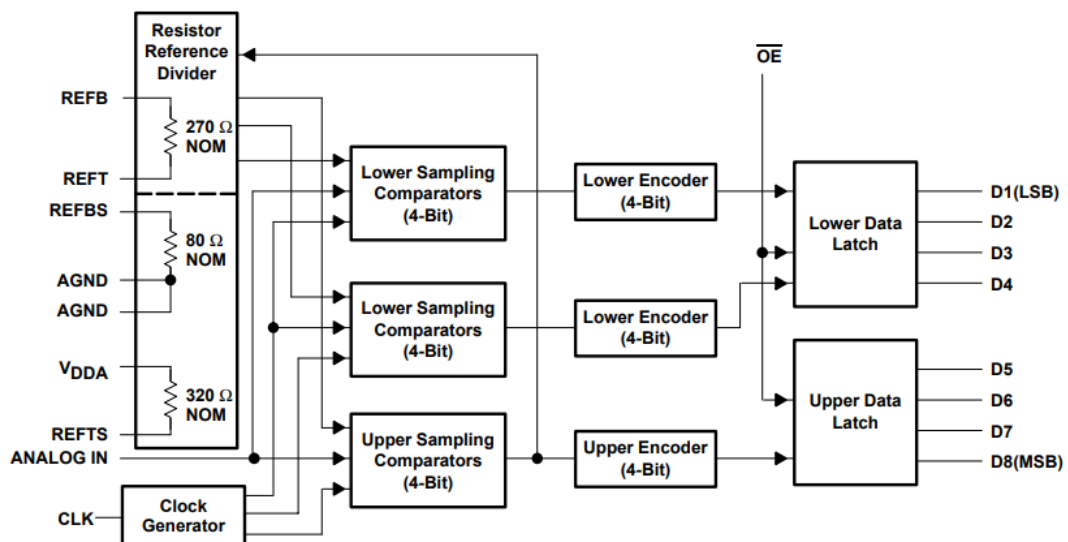


Figure 1.3 TLC5510 Analog-to-digital converter<sup>[15]</sup>

### 1.2.2 Analog-to-Digital Converters (ADCs)

Figure 1.3 shows an 8-bit, analog-to-digital converter (ADCs) that utilizes a semiflash architecture<sup>[15]</sup>. It contains four main parts: resistor reference divider,

comparators, encoders, and data latches. The resistor reference divider part has three sets, a total of 4 reference voltage inputs. These reference voltages will be divided by resistors and introduced to the comparator. After comparison, the result will encode by the encoder and enters the data latch to output the digital signal. The working principle of digital-to-analog converters (DACs) is similar to this. After the digital is input, the voltage division is controlled by the encoder to output the analog signal. It is not difficult to see from its working principle that the dynamic range and conversion error of this type of ADCs and DACs are directly related to the reference voltage.

### 1.3 Why Reference Voltage

So people may ask, why do we have to specially design a reference voltage? Isn't it possible to use the power supply directly?

Circuits require an ideal source that can generate a stable voltage or current that allows circuits to compare with other voltage or current, so the circuit can "know" it. In a voltage regulator, the reference voltage is used to compare with output voltage then use error amplifier and negative feedback to ensure its stability at the specific value. In an ADC, multiple voltage references are needed to let the circuit compare it with the input analog signal and convert it to a digital signal. In a 5V ADC, a reference voltage accuracy of  $\pm 0.1\%$  ( $\pm 5\text{mV}$ ) will give a 10-bit (resolution 4.88mV) conversion accuracy. To achieve a 12-bit accuracy (resolution 1.22mV), the accuracy of reference voltage must be lower than

$\pm 0.02\%$  ( $\pm 1\text{mV}$ ). So the accuracy of the references will directly affect the accuracy of ADCs and DACs. It can be easy to create a stable reference when the external environment is stable. Unfortunately, the current and resistor will generate heat and cause temperature rise and the ambient temperature can affect the properties of components in the circuit. This is critical when designing and constructing circuits and devices and should be considered carefully. The temperature dependence of most conductors is to a linear degree while semiconductor is exponential. Another example, the base-emitter voltage of a BJT transistor is complementary to absolute temperature and has a temperature coefficient of around  $-1.5\text{mV}/^\circ\text{C}$  meaning that when temperature increases from 0 degrees to 100 degrees, the base-emitter voltage can drop up to 200mV. This may cause the whole circuit to malfunction.

## 1.4 Common Voltage Reference Structure

The major goal of a voltage reference is to provide a voltage that is independent of temperature. To reach this goal, one way is to find a voltage that is insensitive or less sensitive to temperature such as diode reference. Another way is to use two complementary voltages and add to get a temperature-independent voltage such as a bandgap reference.

### 1.4.1 Zener-Diode-Based Reference

Zener-diode-Based voltage reference employs the property of Zener diode. A reverse-biased Zener diode exhibits a controlled breakdown and allows the

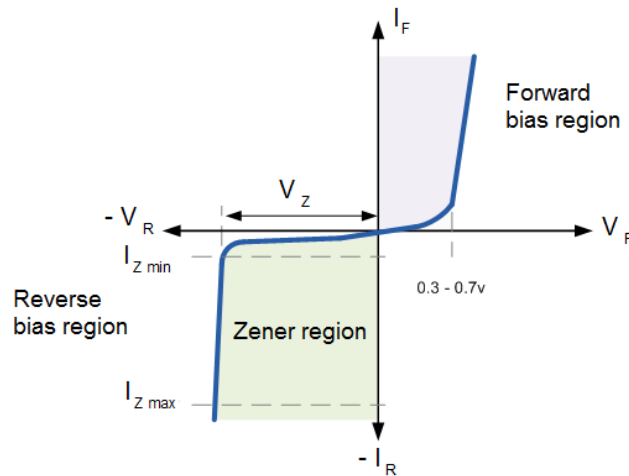


Figure 1.4 Voltage-Current Characteristics of Zener Diode

current to keep the voltage across the Zener diode close to the Zener breakdown voltage, therefore make it ideal for reference voltage. However, the accuracy of reference voltage based on this property is not high. For example, Zener voltage regulator BZX84C2V7LTG1 which is based on this principle, has a nominal reference voltage of 2.7V the maximum is 2.9 and the minimum is 2.5, given it a accuracy of  $\pm 7.4\%$  and makes it only suitable for low accuracy application.

### 1.4.2 Buried Zener Voltage Reference

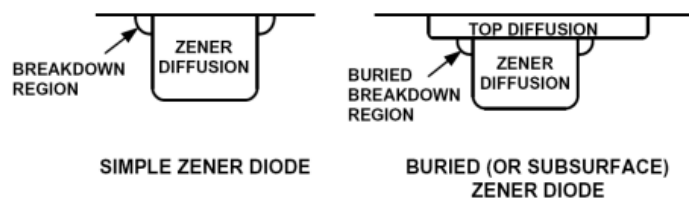


Figure 1.5 Surface Zener diode and buried Zener diode

However, a reference voltage using a buried Zener diode can achieve extremely high accuracy. In an IC chip, diode junctions using surface processes are susceptible to defect. Therefore, the Zener diode formed on the surface may have more noise and become less stable than the embedded Zener diode. This can significantly improve the noise and drift performance compares to surface-type Zener diode<sup>[6]</sup>. This paper<sup>[7]</sup> describes a 5V temperature regulated voltage reference using a buried Zener diode and laser trimmed resistor that can achieve 0.3ppm/°C TC over the temperature range -55°C to 125°C.

The buried Zener diode greatly improved the accuracy of Zener-diode-based voltage reference. Whether it is an ordinary Zener-diode or buried Zener-diode voltage reference, the voltage required for power supply is much higher than 5V and the power consumption is high. The power supply of BZX84C2V7LTG1 is 18V with a power consumption of 250mV and the buried Zener diode voltage reference above is 11.4V ~ 40V with a power consumption of 0.65W at room temperature.

### 1.4.3 Bandgap Reference

A bandgap reference is formed by two parts: a voltage that is proportional to absolute temperature (PTAT) and a voltage that is complementary to absolute temperature (CTAT). The voltage difference between two diodes that operate at different current densities is used to generate a voltage that is proportional to absolute temperature. The complementary to absolute temperature is



generated by a diode operating at a constant current with a TC of approximately  $-1.5\text{mV/K}$ . These two voltages will be added by an operational amplifier. If the ratio of two voltages is chosen properly, the resulting voltage is about  $1.2\text{V}$  with zero TC.

#### 1.4.4 XFET Reference

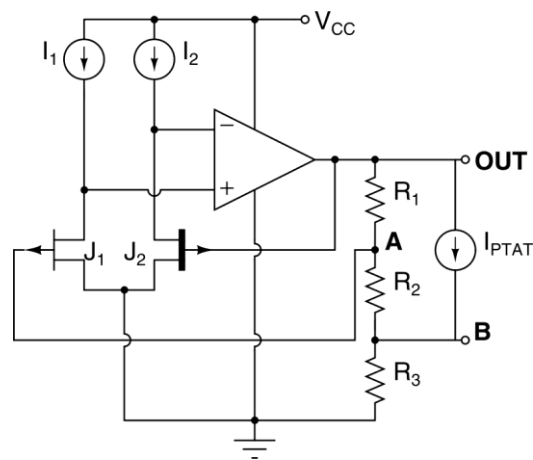


Figure 1.6 Structure of XFET reference<sup>[8]</sup>

The XFET reference is a new reference technique that consists of two junction field-effect transistors, one of which has an extra channel implant to raise the pinch-off voltage. These two JFET form the core of the circuit. As shown in figure 1.6, J1 and J2 are driven with the same current from matched current sources. J2 has an extra channel implant result a pinch-off voltage difference between J1 and J2. When the current and source voltage is the same, a differential voltage will appear between the gates and be amplified by the operational amplifier to form the reference voltage.

#### 1.4.5 Comparison

|                   | Zener diode reference | Buried Zener reference | Bandgap reference     | XFET reference              |
|-------------------|-----------------------|------------------------|-----------------------|-----------------------------|
| Supply voltage    | >5V                   | >5V                    | <5V                   | <5V                         |
| process           | Standard COMS process | Need special process   | Standard COMS process | Additional implant required |
| TC                | High                  | Very Low               | Low                   | Low                         |
| Power dissipation | High                  | High                   | Low                   | Low                         |

Table 1 Reference structure comparison

The buried Zener reference has the lowest TC among those references. But the high supply voltage (>5V) and high-power consumption limit its usage. The bandgap and XFET reference have a low power supply and relatively close accuracy. But XFET reference requires an additional implant in the channel while the bandgap only needs regular MOSFET, bipolar, and resistors. These made the bandgap reference the most widely used reference type in voltage regulators and converters.

## 1.5 Bandgap reference

The first bandgap reference circuit was published by David Hilbiber in 1964<sup>[3]</sup> followed by with commercially successful versions by Bob Widlar<sup>[9]</sup>, Paul Brokaw<sup>[10]</sup>, and others<sup>[13]</sup>. The first IC using the bandgap principle is a 5-volt regulator LM109.

The temperature properties of components bring a lot of trouble in designing the circuit. But we can also use this property to create references that are independent of temperature. As we mentioned before, the base-emitter voltage

of a BJT transistor is complementary to absolute temperature. If we can generate a voltage that is proportional to the absolute temperature, then we may be able to compensate for this base-emitter voltage and get a voltage that is independent of temperature. This forms the core circuit of bandgap reference.

### 1.5.1 Bandgap of semiconductors

In semiconductors and insulators, electrons are confined to bands of energy and forbidden from other regions. The word “bandgap” refers to the gap between the top of the valence band and the bottom of the conduction band. Electrons can jump from one band to another band. When electrons jump from

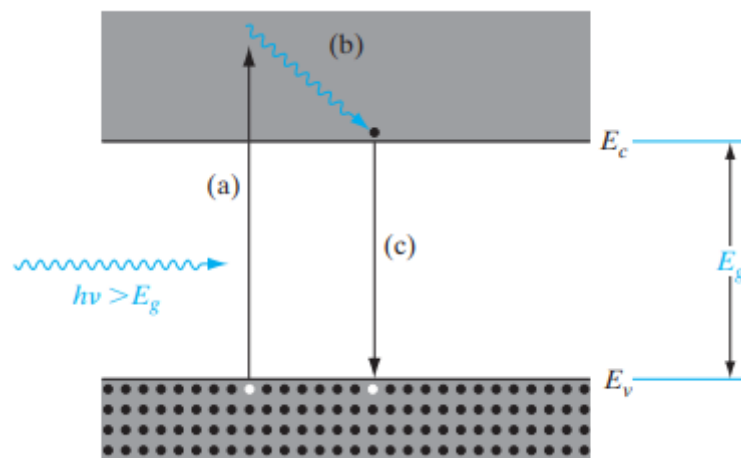


Figure 1.7 Electron move between two bands<sup>[11]</sup>

the conduction band to the valence band, they can release energy in the form of light or heat. When an electron jumps from the valence band to the conduction band, it requires a minimum energy which depends on the bandgap energy.

The bandgap energy of semiconductors tends to decrease when the temperature increases. The relation between bandgap energy and temperature can be described by Varshni's empirical expression:

$$E_g(T) = E_g(0) - \frac{\alpha}{\beta + T} \quad (1.1)$$

Where  $E_g(0)$ ,  $\alpha$ , and  $\beta$  are material constants. For silicon, the  $E_g(0)$  is 1.1557eV,  $\alpha$  is  $7.021 \times 10^{-4}$ ,  $\beta$  is 1108<sup>[12]</sup>.

### 1.5.2 Negative-TC Voltage

The negative-TC can be generated by the base-emitter voltage of BJT transistors, or generally, the forward voltage of a PN-junction diode. Its first derivative with respect to temperature can be expressed as follow:

$$\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4 + m)V_T - \frac{E_g}{q}}{T} \quad (1.2)$$

With  $V_{BE} \approx 750\text{mV}$  and  $T = 300\text{K}$ ,  $\partial V_{BE}/\partial T \approx -1.5\text{mV/K}$ .

### 1.5.3 Positive-TC Voltage

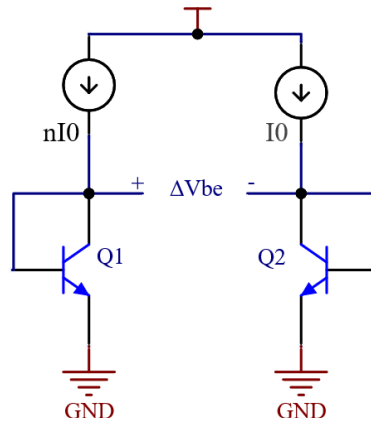


Figure 1.8 positive TC

It was recognized in 1964 that if two BJT transistors operate at unequal current densities, then the difference between their base-emitter voltage is proportional to the absolute temperature. As shown in figure 1.8, if two identical transistors are biased at collector currents of  $nI_0$  and  $I_0$  and their base currents are negligible, then

$$\Delta V_{BE} = V_T \ln n \quad (1.3)$$

Thus

$$\frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q} \ln n \quad (1.4)$$

So the  $V_{BE}$  difference exhibits a positive temperature coefficient.

#### 1.5.4 Supply-Independent Biasing

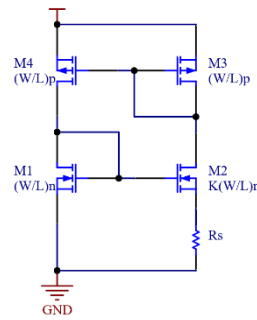


Figure 1.9 Supply independent biasing

When generating negative-TC and positive-TC voltages, we use some current sources and mirrors to provide the bias current. However, the output current of such circuit is quite sensitive to supply voltage. In order to get a less sensitive current, the circuit must bias itself. As shown in figure 1.9, M3 and M4 copy  $I_{out}$  and thereby define  $I_{REF}$ . Here we have

$$I_{out} = \frac{2}{\mu_n C_{ox} \left(\frac{W}{L}\right)_N} \cdot \frac{1}{R_S^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \quad (1.5)$$

Which is independent of the supply voltage.

### 1.5.5 Bandgap Reference Voltage

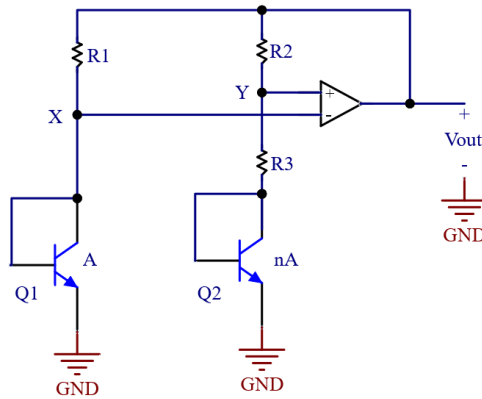


Figure 1.10 Bandgap reference

Figure 1.10 shows a simple bandgap reference. Amplifier A<sub>1</sub> senses V<sub>X</sub> and V<sub>Y</sub>, driving the top terminals of R<sub>1</sub> and R<sub>2</sub> such that V<sub>X</sub> ≈ V<sub>Y</sub>. Here we have V<sub>BE1</sub> – V<sub>BE2</sub> = V<sub>T</sub>lnn. Therefore, V<sub>OUT</sub> = V<sub>BE2</sub> + (V<sub>T</sub>lnn) (1 +  $\frac{R_2}{R_3}$ ). For a zero TC, we must have  $\ln n \left(1 + \frac{R_2}{R_3}\right) \approx 17.2$  [14]. We can notice that the result is not dependent on the TC of resistors.

### 1.5.6 Bandgap Reference Current

When analyzing the bandgap circuit, we can find that the bias current of the bipolar transistors is proportional to absolute temperature. So the PTAT current can be generated by mirroring the current through bipolar.

## 1.6 Research Objectives

Conventional bandgap references produce reference voltage close to 1.2V meaning that the supply voltage must be greater than 1.4V. This limits the application in the sub-1V system. Also, the power consumption of conventional bandgap reference is usually a few hundred microwatts to milliwatts. This also limits their use in low-power devices.

This research project is developing an ultra-low-power ( $<1\mu\text{W}$ ) all-in-one bandgap reference suitable for low power applications to provide more application compare to conventional bandgap reference.

## 2 Literature Review

Since its first introduction by Widlar<sup>[3]</sup>, the bandgap referencing (BGR) technique has been widely employed for implementing a voltage reference source in bipolar integrated circuits<sup>[17]</sup>. The temperature stability of the bandgap reference has been continuously improved via new structure, new technology, and new process such as curvature compensation and laser trim<sup>[18]-[21]</sup>. Meanwhile, the development of IoT devices and wearable devices also increases the demand for low power consumption.

### 2.1 Bandgap Reference Voltage

#### 2.1.1 Conventional Bandgap Reference Structure

Akshaya et al.<sup>[22]</sup> describe a bandgap reference circuit that can achieve very high accuracy (0.2ppm) and low power consumption with a conventional bandgap structure. To achieve such accuracy, they tried to compensate the linear and curvature components shown in the Taylor series expansion of logarithmic term, which can result in temperature coefficient ranging from 50 to



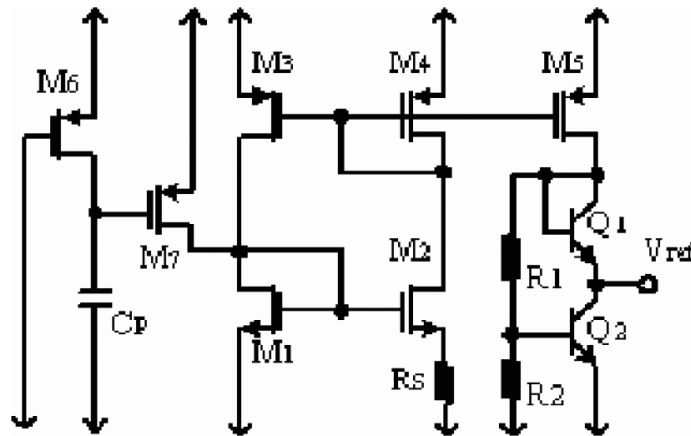


The conventional bandgap reference obtains the reference voltage by adding PTAT (proportional to absolute temperature) and CTAT (complementary to absolute temperature) voltage which limits the minimum reference voltage. To get a lower reference voltage, an alternative solution is to sum the CTAT and PTAT current. Banba et al.<sup>[25]</sup> describe a novel bandgap reference structure where the reference voltage is obtained from the sum of two currents, one proportional to the diode  $V_f$  and the other proportional to the thermal voltage  $V_T$ . By using this structure, the reference voltage can be set at any level between 0V and  $V_{cc}$  and the supply voltage can be as low as 0.84V in the simulation. This design also has a good low power performance. The operation current is 2.2uA. However, in order to obtain low power consumption and accuracy, a large number of resistors and transistors are used, resulting in excessive chip area.

### 2.1.3 Bandgap Reference Employ Subthreshold Feature

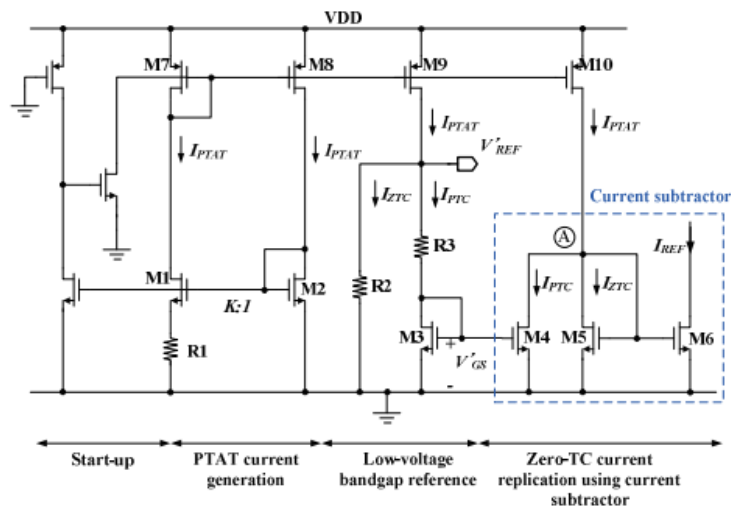
Another way to achieve a low power consumption is to employ the subthreshold characteristics of the MOSFET.

a.

Figure 2.3 Structure of the circuit<sup>[24]</sup>

Lin et al.<sup>[24]</sup> describe a circuit employing this feature that achieves a minimum supply voltage of 0.95V and 2.4uW power consumption. But the reference voltage is only 170mV. This reference voltage is too low and not easy to use.

b.

Figure 2.4 Structure of the circuit<sup>[26]</sup>

Pack et al.<sup>[26]</sup> describe a low-power bandgap reference circuit that employ the subthreshold feature of the MOSFET and reduces the supply voltage by adding

two currents instead of voltage. This circuit also provides a reference current by using a current subtractor. This circuit can provide a 512mV, 54ppm/°C reference voltage and a 0.94uA reference current with a minimum supply voltage of 0.7V and has a total current consumption of 7.6uW.

C.

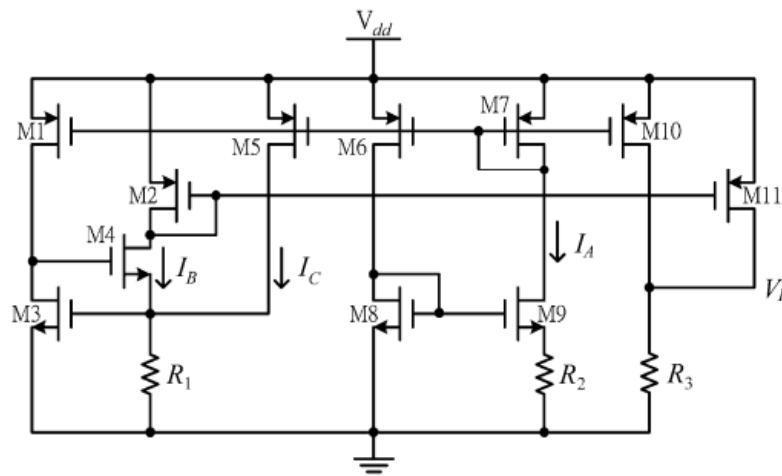
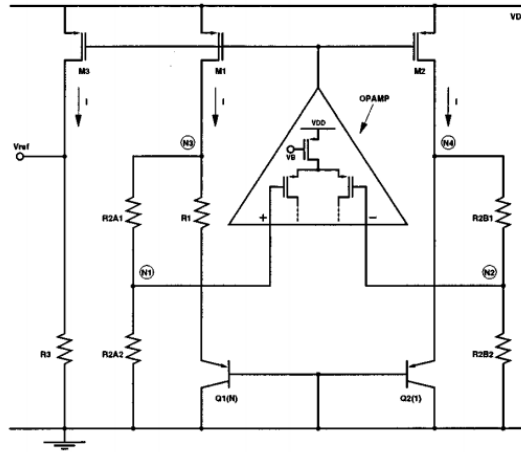


Figure 2.5 Structure of the circuit<sup>[27]</sup>

Huang et al.<sup>[27]</sup> describe a bandgap reference circuit that compensates for the channel-length modulation. As shown in figure 2.5, current  $I_C$  mirroring  $N$  times of  $I_A$  and a suitable value for  $N$  can reduce the channel-length modulation<sup>[27]</sup>. With this compensation, this circuit can provide a 221mV reference voltage with an average power dissipation of 3.3uW.

d.

Figure 2.6 Structure of the circuit<sup>[28]</sup>

Usually, a bandgap reference employing the subthreshold feature will need a low threshold voltage device to reduce the supply voltage. However, Leung and mok<sup>[28]</sup> describe a sub-1-V 15- ppm/°C bandgap voltage reference without requiring a low threshold voltage device. It has a minimum supply voltage of 0.98V with  $V_{thn} \approx |V_{thp}| \approx 0.9V$  and the supply voltage can be reduced to 0.75V when low threshold voltage P-MOS transistors are used.

## 2.2 Comparison

The conventional bandgap reference shown in the first paper usually has a high accuracy reference voltage of 1.2V requiring a power supply greater than 1.5V, increasing the power consumption and limiting the application in low-power and sub-1V systems. In the second structure, the sum of the current replaces the conventional voltage summation which results in a low power supply and low power consumption, but requires many resistors and transistors increasing the

chip area. The third and fifth papers describe a reference circuit that employs the subthreshold characteristic features and achieves low power supply and low power consumption. But the reference voltage is too low making it hard to use.

|                   | [22]      | [25]   | [24]       | [26]     | [27]   | [28]     |
|-------------------|-----------|--------|------------|----------|--------|----------|
| Supply voltage    | 1.8V-3.3V | >0.84V | 0.95V-2.5V | >0.7V    | >0.85V | >0.98V   |
| Reference voltage | 1.12V     | 518mV  | 170mV      | 512mV    | 222mV  | 603mV    |
| Accuracy          | 0.2ppm/°C | ---    | 0.76%      | 54ppm/°C | ---    | 15ppm/°C |
| Power consumption | 52.37uW   | 4.8uW  | 2.4uW      | 5.32uW   | 3.3uW  | 17.64uW  |

Table 2 Circuit comparison

As shown in table 2, the second and third designs are suitable for low-power applications. Although the reference voltage of the second design is competitive with the third, the structure of the second design is much more complicated than the third one. The lower output voltage of the third design can be improved with some minor modifications giving it great potential for low power applications.

Based on the analysis, this research proposes an ultra-low-power (<1uW) bandgap reference circuit with a high reference voltage (>0.7V). In order to make this circuit low power, the subthreshold feature of the MOSFET must be used to reduce the current. Meanwhile, to reduce the supply voltage, the conventional voltage-based bandgap structure will be replaced by a novel current-based structure.

## 3 Bandgap Reference Design

### 3.1 Bandgap reference core circuit

#### 3.1.1 Voltage part

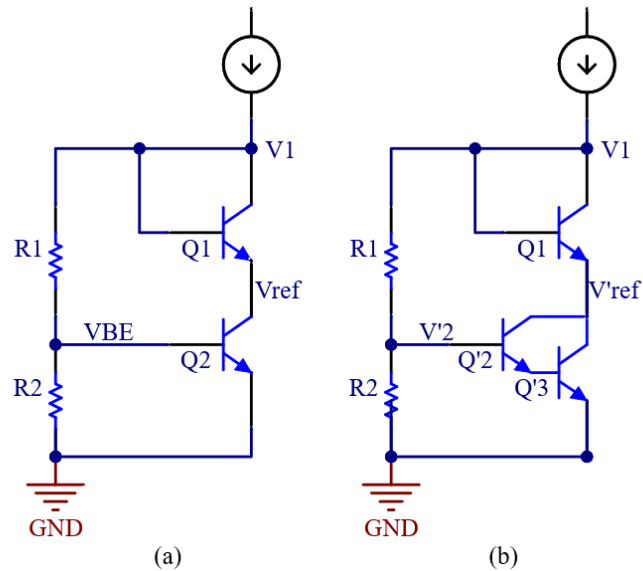


Figure 3.1 BGR voltage core circuit

Fig 3.1(a) shows a simplified BGR core circuit. The area ratio of NPN transistors  $Q_1$  to  $Q_2$  is  $N$ . If the base current of  $Q_2$  is much smaller than the current through  $R_1$  and  $R_2$ , the node voltage  $V_1$  can be written as

$$V_1 = V_{BE2} + \frac{R_1}{R_2} V_{BE2} \quad (3.1)$$

So  $V_1$  is proportional to  $V_{BE2}$  which has a negative TC. If  $V_1$  is subtracted from

$V_{BE1}$  the reference voltage  $V_{ref}$  can be expressed as

$$\begin{aligned}
 V_{ref} &= V_1 - V_{BE1} \\
 &= V_{BE2} - V_{BE1} + \frac{R_1}{R_2} V_{BE2} \\
 &= \Delta V_{BE} + \frac{R_1}{R_2} V_{BE2}
 \end{aligned} \tag{3.2}$$

If  $Q_1$  is diode-connected and the ratio ( $\beta$ ) of the collector current ( $I_{C2}$ ) to the base current ( $I_{B2}$ ) of  $Q_2$  is large enough, the base current of  $Q_2$  can be ignored, so  $I_{E1} = I_{C2}$  and we have

$$\begin{aligned}
 \Delta V_{BE} &= V_{BE2} - V_{BE1} \\
 &= V_T \ln \frac{I_0}{I_s} - V_T \ln \frac{I_0}{NI_s} \\
 &= V_T \ln N \\
 &= \frac{KT}{q} \ln N
 \end{aligned} \tag{3.3}$$

After the equation 3-3 is substituted into equation 3-2, the reference voltage is obtained.

$$V_{ref} = \frac{KT}{q} \ln N + \frac{R_1}{R_2} V_{BE2} \tag{3.4}$$

By changing the ratio of  $R_1$  and  $R_2$  we can get a lower reference voltage compared to conventional bandgap voltage which is around 1.25V.

However, the  $R_2$  here is very large to reduce the power consumption so the ratio is usually very low. This leads to a very low reference voltage that is usually around 200mV<sup>[24] [27]</sup> which is hard to use. In order to increase the reference voltage without increasing the resistance area, a Darlington



structure was employed as shown in fig 3.1b.  $Q_2$  was replaced with  $Q'_2$  and  $Q'_3$  in a Darlington structure. The area ratio of NPN transistors  $Q_1$  to  $Q_3$  is  $N_1$ ,  $Q_2$  to  $Q_3$  is  $N_2$ . The  $V_{BE2}$  in 3.1a is changed to  $V'_2 = V'_{BE2} + V'_{BE3}$ , so  $V_1 = V_2 + \frac{R1}{R2}V_2$ , therefore we can rewrite equation 3-2 as

$$\begin{aligned} V_{ref} &= V_1 - V_{BE1} \\ &= V'_{BE2} + V'_{BE3} - V_{BE1} + \frac{R1}{R2}(V'_{BE2} + V'_{BE3}) \end{aligned} \quad (3.5)$$

In the Darlington structure, the current through  $Q'_3$  ( $I'_{C3}$ ) is  $\beta$  times that of  $Q'_2$  ( $I'_{C2}$ ) and if the  $\beta$  is larger enough, the current of  $Q'_2$  can be ignored. So  $I'_{C2} \approx 0$  and  $I'_{E1} = I'_{C3}$ . So we have

$$\begin{aligned} V'_{BE2} + V'_{BE3} - V_{BE1} &= V_T \ln \frac{0}{I_s} + V_T \ln \frac{I_0}{N_2 I_s} - V_T \ln \frac{I_0}{N_1 I_s} \\ &= V_T \left( 1 + \ln \frac{N_1}{N_2} \right) \\ &= \frac{KT}{q} \left( 1 + \ln \frac{N_1}{N_2} \right) \end{aligned} \quad (3.6)$$

After the equation 3.6 is substituted into equation 3.5, the reference voltage is obtained.

$$V_{ref} = \frac{KT}{q} \left( 1 + \ln \frac{N_1}{N_2} \right) + \frac{R1}{R2} (V'_{BE2} + V'_{BE3}) \quad (3.7)$$

Comparing equations 3.7 and 3.4, we can find that by using the Darlington structure, the reference voltage is increased by  $\frac{KT}{q} \left( 1 + \ln \frac{N_1}{N_2} - \ln N \right) + \frac{R1}{R2} V'_{BE3}$ , if we take  $N_1 = 8$ ,  $N_2 = 2$ ,  $N = 8$ , the voltage increased  $0.3 \frac{KT}{q} + \frac{R1}{R2} V'_{BE3}$ , which is more than twice the original reference voltage. Meanwhile, the current through  $Q_2$  decreases and therefore improves the accuracy. Also,

the power consumption is almost the same since we didn't change the resistor (R1 and R2) and the current of  $Q'_2$  can be ignored.

### 3.1.2 Current part

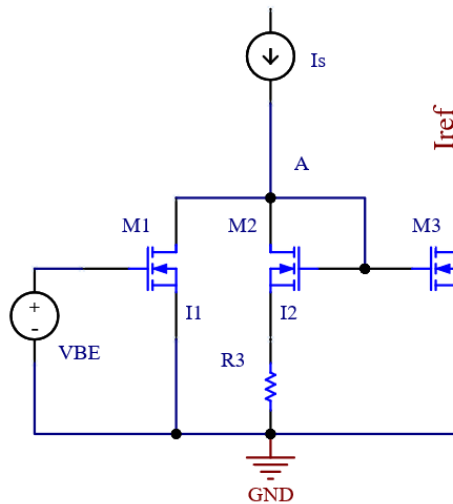


Figure 3.2 BGR current core circuit

Fig 3.2 shows the BGR current core circuit. The current source is proportional to absolute temperature. The W/L ratio of M1, M2, and M3 is  $W/L_1$ ,  $W/L_2$ , and  $W/L_3$ . M1 and M2 are in weak inversion, M3 is in saturation. Apply KCL at A point, we have

$$I_s = I_2 + I_1 \quad (3.8)$$

therefore  $I_2 = I_s - I_1$ .  $I_s$  is the current source which can be represent to

$$I_s = C_1 T \quad (3.9)$$

Where C is a constant. For MOSFET in weak inversion, we have

$$I_D = I_{D0} \frac{W}{L} e^{\frac{q(V_{GS})}{nkT}} \quad (3.10)$$

Employ Taylor Series here,  $e^x = 1 + x + \frac{x^2}{2!} + \dots$  (for all x), we have

$$I_1 = I_{D0} \left( \frac{W}{L} \right)_1 \left( 1 + \frac{q(V_{GS})}{nkT} \right) \quad (3.11)$$

Where  $V_{GS} = V_{BE} = V_{BE0} - 0.0015T$ . So  $I_1$  can be written as

$$I_1 = I_{D0} \left( \frac{W}{L} \right)_1 - \frac{0.0015q}{nk} + \frac{qV_{BE0}}{nkT} \quad (3.12)$$

Bring  $I_1$  into  $I_2$  we have

$$I_2 = C_1 T - \frac{qV_{BE0}}{nkT} - I_{D0} \left( \frac{W}{L} \right)_1 + \frac{0.0015q}{nk} \quad (3.13)$$

Let  $C_2 = \frac{qV_{BE0}}{nk}$ ,  $C_3 = \frac{0.0015q}{nk}$

$$I_2 = C_1 T - C_2 \frac{1}{T} + C_3 - I_{D0} \left( \frac{W}{L} \right)_1 \quad (3.14)$$

M2 is in saturation, therefore

$$I_2 = \frac{\mu_n C_{OX}}{2} \left( \frac{W}{L} \right)_2 (V_{GS2} - V_{th})^2 \quad (3.15)$$

$$V_{GS2} = \sqrt{\frac{2}{\mu_n C_{OX}} \left( \frac{L}{W} \right)_2 I_2} + V_{th} \quad (3.16)$$

For M3, we have

$$V_{GS3} = V_{GS2} + I_2 R_1 \quad (3.17)$$

Since M3 is in saturation,

$$I_3 = \frac{\mu_n C_{OX}}{2} \left( \frac{W}{L} \right)_3 (V_{GS3} - V_{th})^2 \quad (3.18)$$

Bring  $V_{GS3}$  into  $I_3$ , we have

$$I_3 = \frac{\mu_n C_{OX}}{2} \left( \frac{W}{L} \right)_3 \left( \frac{2}{\mu_n C_{OX}} \left( \frac{L}{W} \right)_2 I_2 + I_2^2 R_1^2 + 2R_1 \sqrt{\frac{2}{\mu_n C_{OX}} \left( \frac{L}{W} \right)_2} I_2^3 \right) \quad (3.19)$$

Let  $C_4 = \frac{\mu_n C_{OX}}{2}$ , for  $I_8^{1.5}$  the Taylor series approximation is  $1 + \frac{3}{2}(x - 1)$  so we have

$$I_3 = C_4 \left( \frac{W}{L} \right)_3 \left( \frac{1}{C_4} \left( \frac{L}{W} \right)_2 I_2 + I_2^2 R_1^2 + R_1 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} (3I_2 - 1) \right) \quad (3.20)$$

Expanding this equation gives Iref

$$\begin{aligned} I_{ref} &= C_1^2 C_4 R_1^2 \left( \frac{W}{L} \right)_3 T^2 \\ &+ \left[ 2C_1 C_3 C_4 R_1^2 \left( \frac{W}{L} \right)_3 + \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_1 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) C_1 \right] T \\ &- \left[ 2C_2 C_3 C_4 R_1^2 \left( \frac{W}{L} \right)_3 + \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_1 C_2 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) \right] \frac{1}{T} \\ &+ C_4 R_1^2 \left( \frac{W}{L} \right)_3 C_2^2 \frac{1}{T^2} \\ &+ C_4 R_1^2 \left( \frac{W}{L} \right)_3 (-2C_1 C_2 + C_3^2) + C_3 \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_1 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) - R_1 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \quad (3.21) \end{aligned}$$

Differentiating gives

$$\begin{aligned} \frac{\partial I_{ref}}{\partial T} &= 2C_1^2 C_4 R_1^2 \left( \frac{W}{L} \right)_3 T \\ &+ \left[ 2C_1 C_3 C_4 R_1^2 \left( \frac{W}{L} \right)_3 + \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_1 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) C_1 \right] \end{aligned}$$

$$\begin{aligned}
& + \left[ 2C_2C_3C_4R_1^2 \left( \frac{W}{L} \right)_3 + \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_1C_2 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) \right] \frac{1}{T^2} \\
& + C_4R_1^2 \left( \frac{W}{L} \right)_3 C_2^2 \frac{1}{T^3}
\end{aligned} \tag{3.22}$$

Ignoring the high-order terms gives

$$\begin{aligned}
\frac{\partial I_{ref}}{\partial T} &= 2C_1^2C_4R_1^2 \left( \frac{W}{L} \right)_3 T \\
& + \left[ 2C_1C_3C_4R_1^2 \left( \frac{W}{L} \right)_3 + C_1 \left( \frac{\left( \frac{W}{L} \right)_3}{\left( \frac{W}{L} \right)_2} + 3R_2 \sqrt{\frac{1}{C_4} \left( \frac{L}{W} \right)_2} \right) \right]
\end{aligned} \tag{3.23}$$

The first derivative shows that the curve is parabolic. Since our working range is from 0 degrees to 100 degrees. In order to minimize the current change in this region, the first derivative must be close to zero, and when the first derivative is equal to zero the temperature should be in the mid-point of the working region

### 3.2 Complete Reference Circuit.

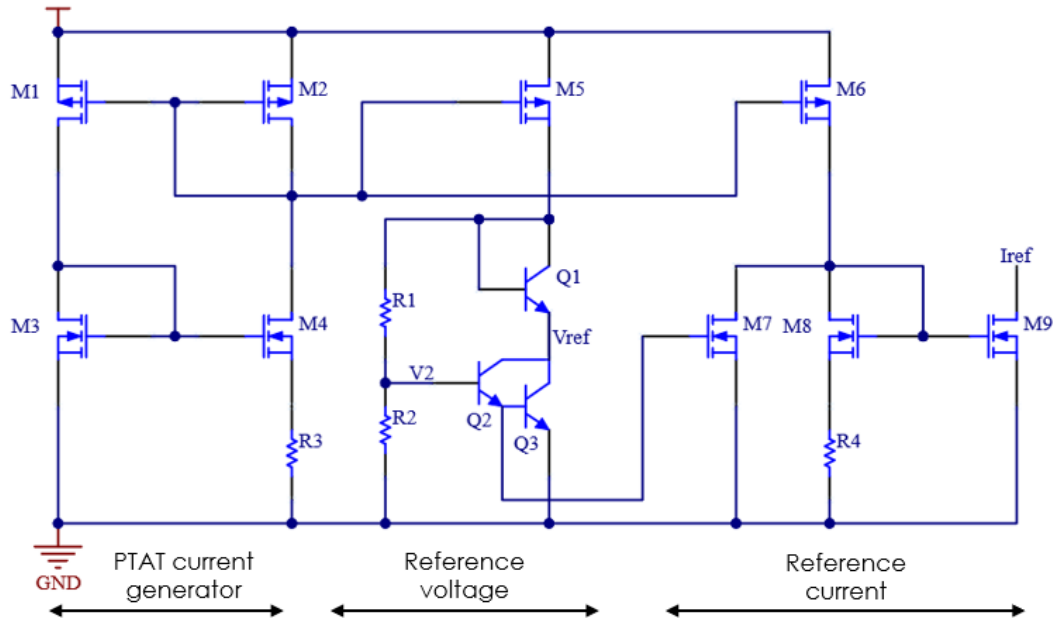


Figure 3.3 Complete Bandgap Reference Circuit

The complete voltage reference circuit shown in Fig 3.2 includes the subthreshold current generator composed of M<sub>1</sub> to M<sub>4</sub> and R<sub>3</sub> and the core circuit utilizing M<sub>5</sub>, R<sub>1</sub>, R<sub>2</sub>, and Q<sub>1</sub> to Q<sub>3</sub>.

The working region of the N-MOSFET in the subthreshold current generator is decided by the current through M<sub>3</sub> and M<sub>4</sub>. Smaller currents can make them work in weak inversion and the drain current can be expressed as<sup>[29]</sup>

$$V_{GS} = n \frac{KT}{q} \ln \frac{I_D}{\left(\frac{W}{L}\right)_4 I_{D0}} \quad (3.24)$$

So the voltage drop across R<sub>3</sub> is

$$V_{R3} = \Delta V_{GS} = V_{GS3} - V_{GS4} = n \frac{KT}{q} \ln \left[ \frac{\left(\frac{W}{L}\right)_4 I_3}{\left(\frac{W}{L}\right)_3 I_4} \right] \quad (3.25)$$

where  $n = 1 + C_d/C_{ox}$ ,  $C_d$  is the surface depletion capacitance,  $C_{ox}$  is the gate oxide capacitance. If  $I_3$  is chosen to be equal to  $I_4$  by using current mirrors of  $M_1$  and  $M_2$ , the current through  $R_3$  can be written as

$$I_3 = \frac{V_{RS}}{R_s} = \frac{n}{R_3} \frac{KT}{q} \ln \left[ \frac{\left(\frac{W}{L}\right)_4}{\left(\frac{W}{L}\right)_3} \right] \quad (3.26)$$

Therefore, the currents through  $M_4$  and  $M_3$  are independent of supply voltage.

Bring  $I_3$  into  $I_{ref}$  formula, we have

$$\begin{aligned} I_{ref} = & C_1^2 C_4 R_4^2 \left(\frac{W}{L}\right)_9 \left(\frac{W}{L}\right)_6^2 T^2 \\ & + \left[ 2C_1 C_3 C_4 R_4^2 \left(\frac{W}{L}\right)_9 \left(\frac{W}{L}\right)_6 + \left( \frac{\left(\frac{W}{L}\right)_9}{\left(\frac{W}{L}\right)_8} + 3R_4 \sqrt{\frac{1}{C_4} \left(\frac{L}{W}\right)_8} \right) C_1 \left(\frac{W}{L}\right)_6 \right] T \\ & - \left[ 2C_2 C_3 C_4 R_4^2 \left(\frac{W}{L}\right)_9 + \left( \frac{\left(\frac{W}{L}\right)_9}{\left(\frac{W}{L}\right)_8} + 3R_4 C_2 \sqrt{\frac{1}{C_4} \left(\frac{L}{W}\right)_8} \right) \right] \frac{1}{T} \\ & + C_4 R_4^2 \left(\frac{W}{L}\right)_9 C_2^2 \frac{1}{T^2} \\ & + C_4 R_4^2 \left(\frac{W}{L}\right)_9 \left( -2C_1 C_2 \left(\frac{W}{L}\right)_6 + C_3^2 \right) + C_3 \left( \frac{\left(\frac{W}{L}\right)_9}{\left(\frac{W}{L}\right)_8} + 3R_4 \sqrt{\frac{1}{C_4} \left(\frac{L}{W}\right)_8} \right) \\ & - R_4 \sqrt{\frac{1}{C_4} \left(\frac{L}{W}\right)_8} \end{aligned} \quad (3.27)$$

$$V_{ref} = \frac{KT}{q} \left( 1 + \ln \frac{N_1}{N_2} \right) + \frac{R1}{R2} (V'_{BE2} + V'_{BE3}) \quad (3.28)$$

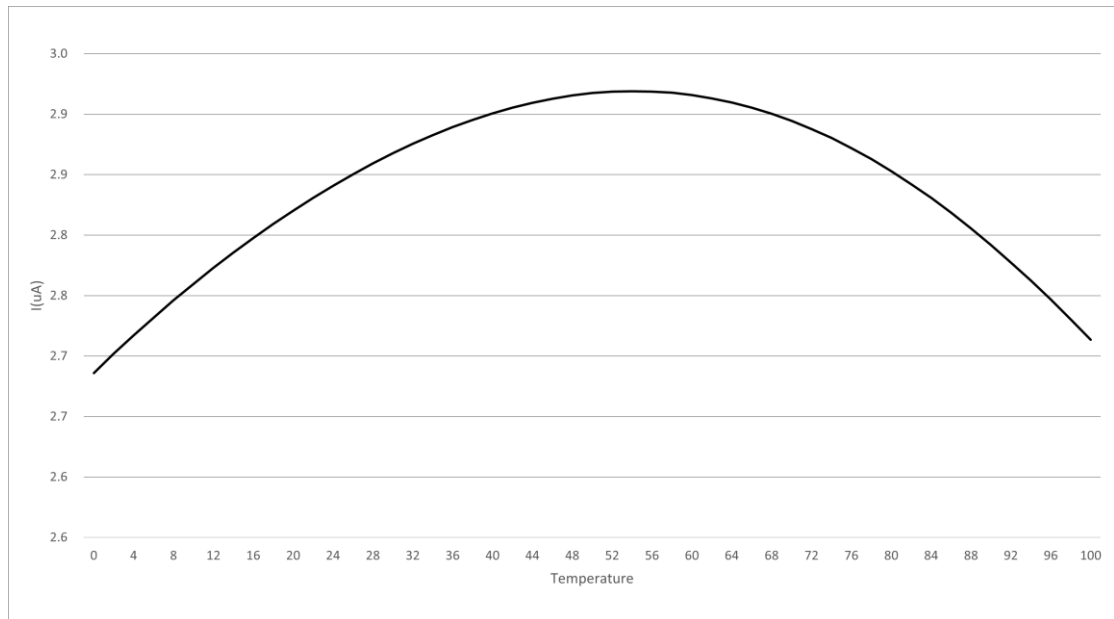


Figure 3.4  $I_{ref}$  formula simulation result



## 4 Simulation Results

The proposed bandgap voltage and current reference are designed with a standard 65nm CMOS process. The simulation result of the temperature coefficient is shown in figure 4.1. The temperature coefficient of the reference voltage and reference current is 63.5ppm/°C and 246.31ppm/°C with the temperature ranging from 0°C to 100°C.

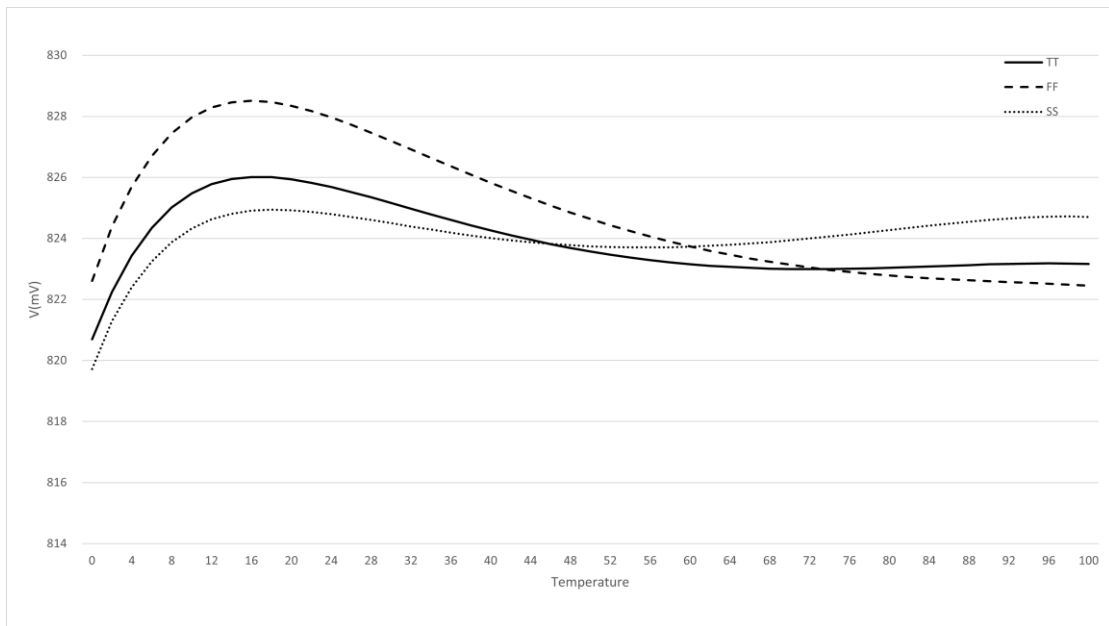


Figure 4.1 Reference voltage simulation

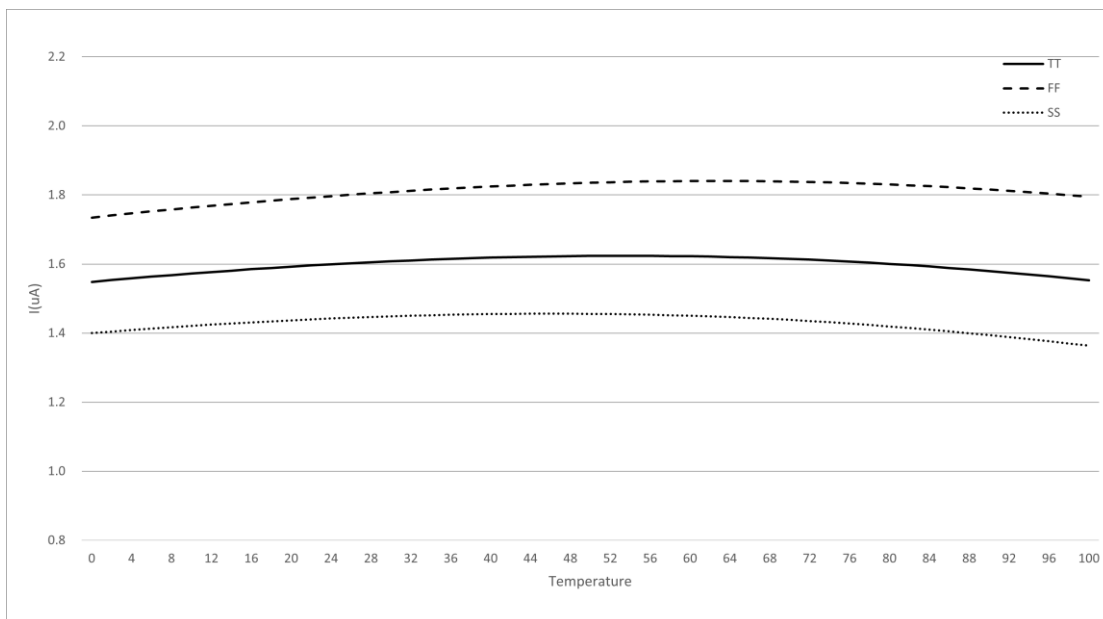


Figure 4.3 Reference current simulation

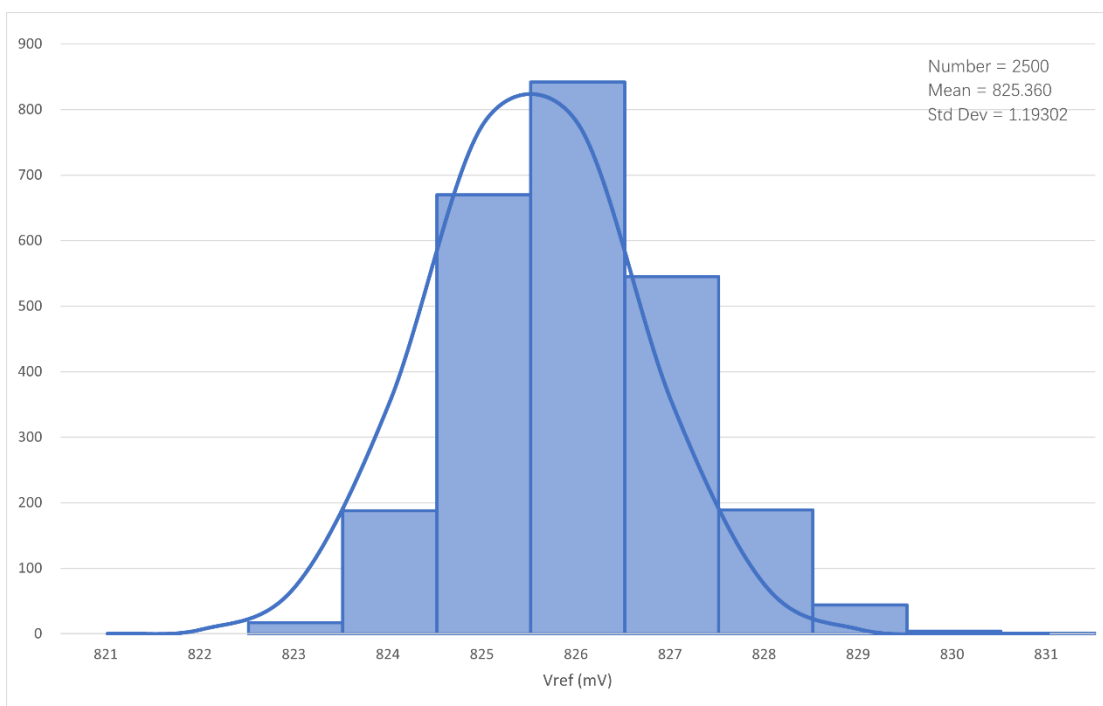


Figure 4.2 2500-run Monte-Carlo Simulation of reference voltage

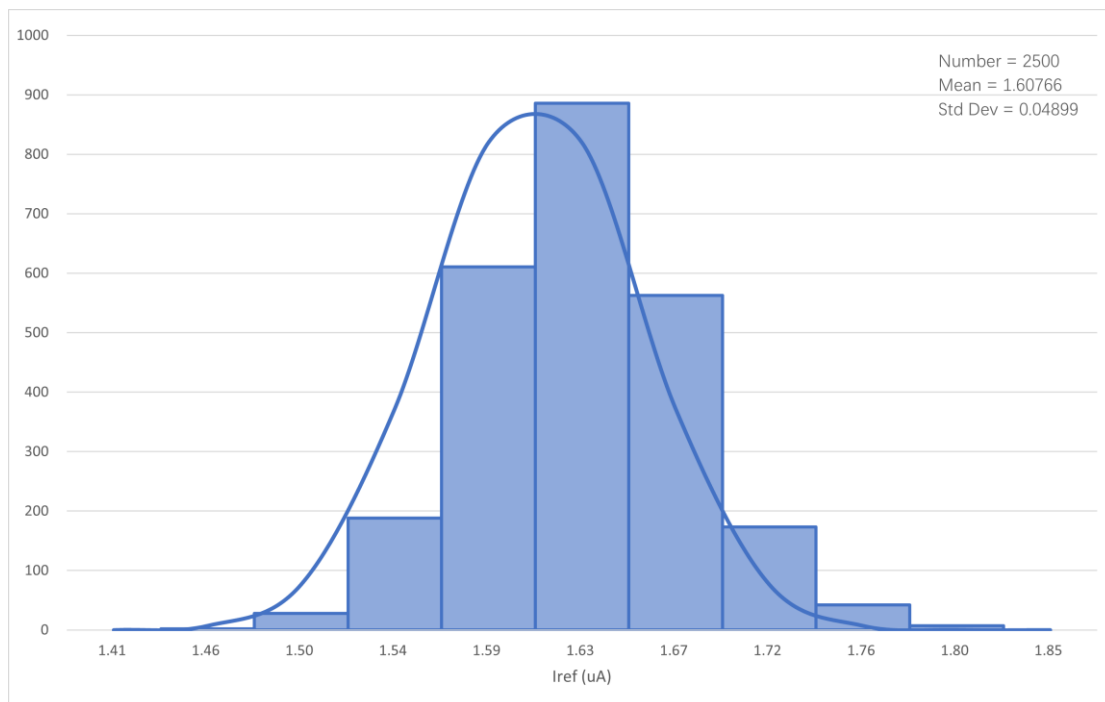


Figure 4.5 2500-run Monte-Carlo Simulation of reference current

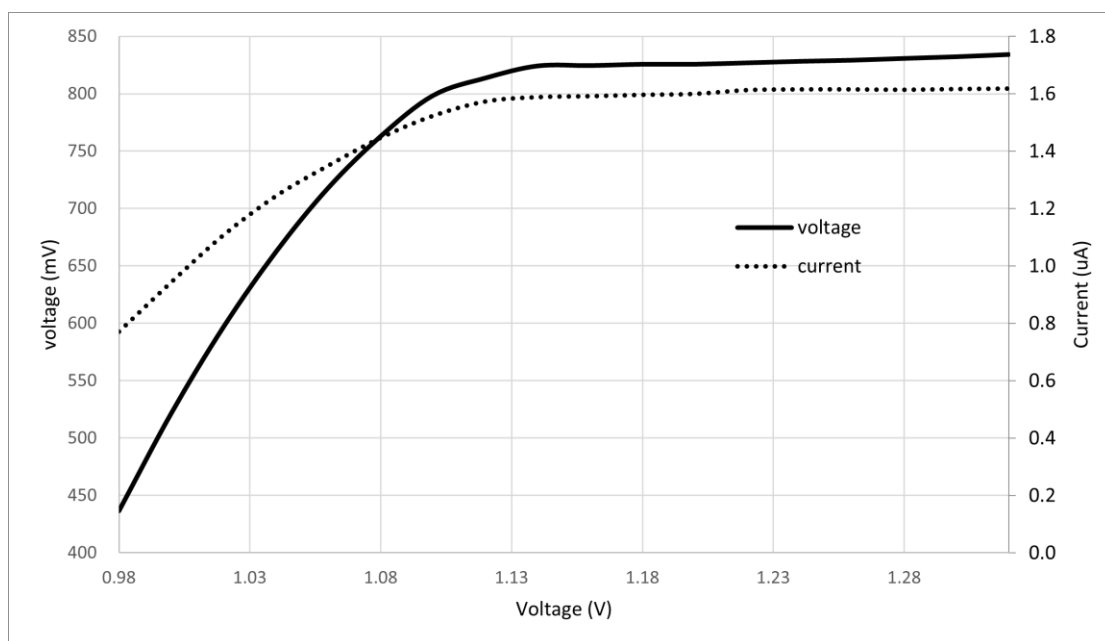


Figure 4.4 Variation with supply

## 5 Measurement Results

Seven chips been tested. At room temperature, the average output voltage is 585mv, the maximum and minimum output voltages are 605mV and 586mV, and the standard deviation is 10.47mV. The chip consumes 810nA at room

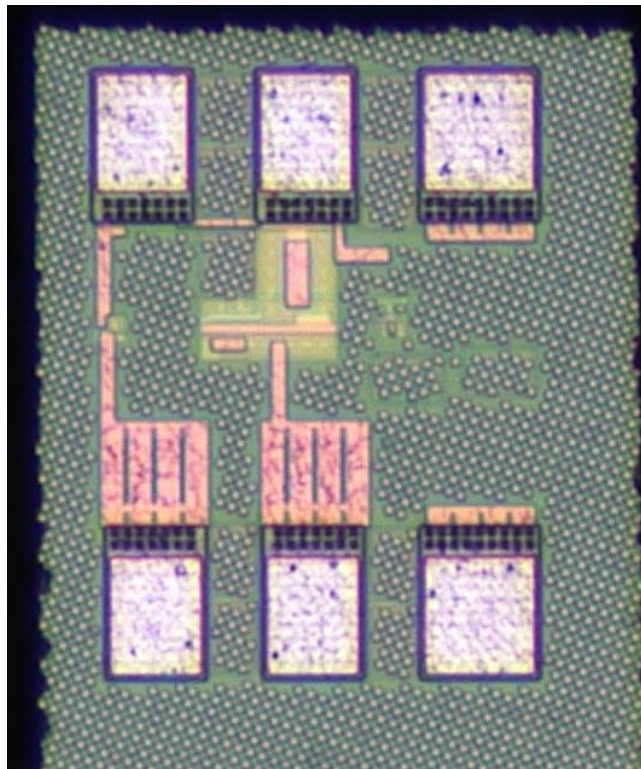


Figure 5.1 Chip photo

temperature which gives it a 972nW power consumption.

The chip has a 3.82% measurement variation and 3.61% fitting curve variation with a 1.6V supply voltage and operating from 0 degrees to 70 degrees.

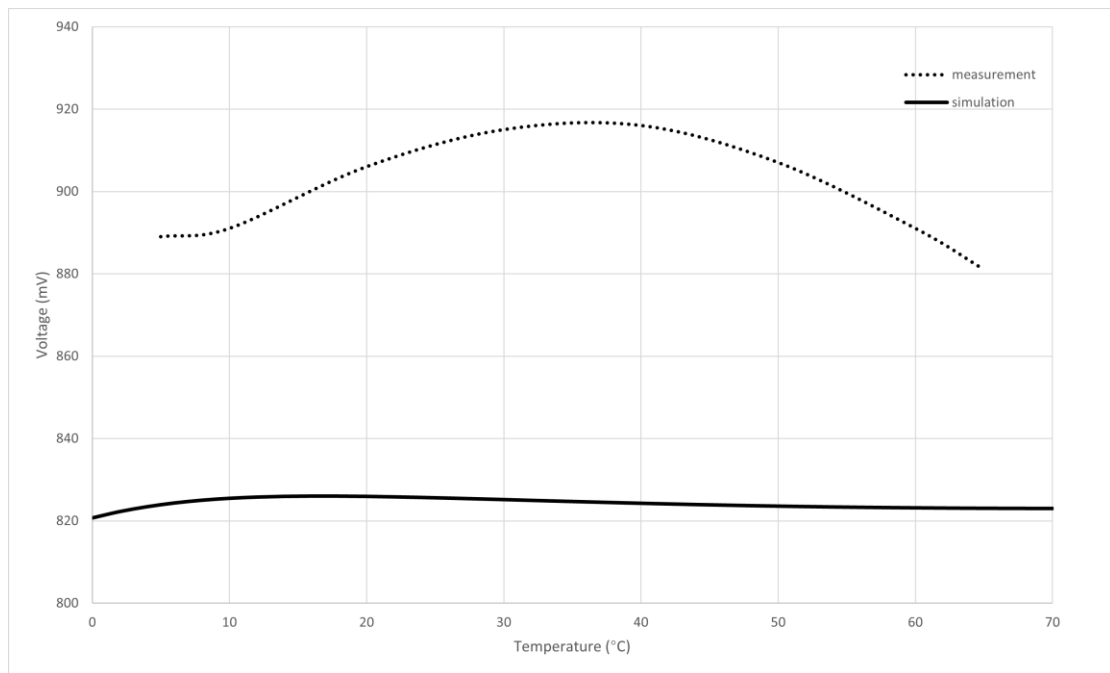


Figure 5.2 Measurement & simulation comparison of reference voltage

There is a big discrepancy between the measured result and the simulated result. A reasonable conjecture is that the lack of doping of the bipolar makes the bipolar not working in near saturation region therefore requiring a higher supply voltage. This also lead to the resistor not match with bipolar leading to large variation.

## 6 Conclusion and Future Work

### 6.1 Conclusion

In this thesis, we developed a new low-power bandgap voltage and current reference. The main highlight of our work is to greatly reduce the power consumption ( $<1\mu\text{W}$ ) of the bandgap voltage reference while maintaining a relatively good accuracy which makes this bandgap reference a suitable candidate for ultra-low power devices. At the same time, we provide a reference current that occupies a very small chip area, which is not available in most voltage references.

### 6.2 Future Work

After the chip was taped out, we further reduced the power consumption of the voltage reference part to  $78\text{nW}$  and increased the accuracy of both the reference voltage and current to  $31/92\text{ ppm}/^\circ\text{C}$ . This shows that this simple

structure has the potential for lower power consumption and higher accuracy.

However, this structure requires large resistors and may cost more chip area.

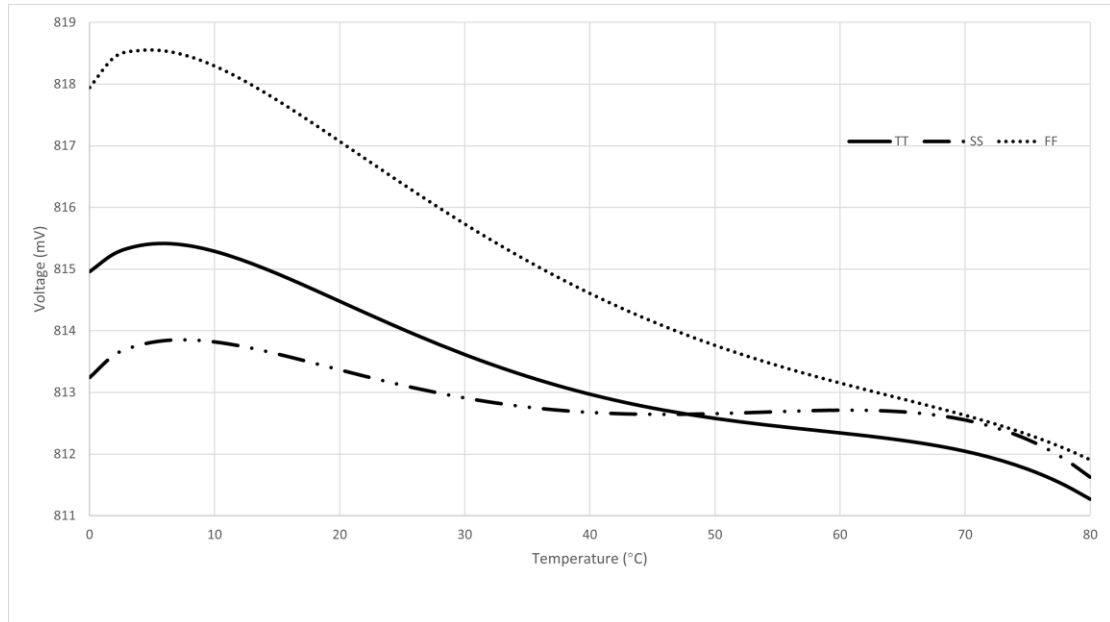


Figure 6.1 Improved voltage reference

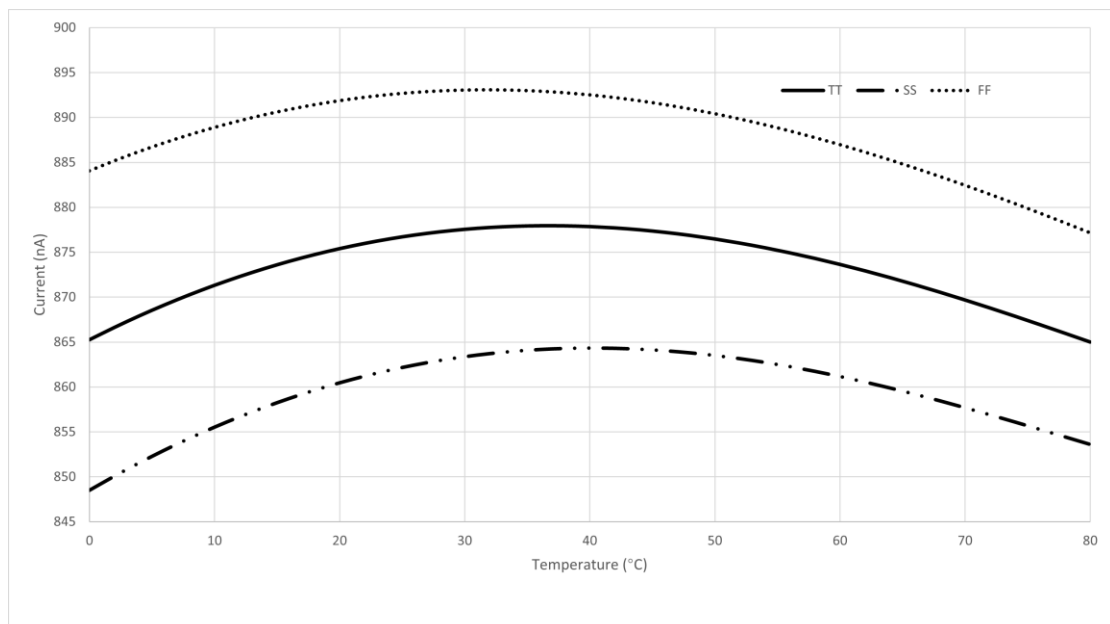


Figure 6.2 Improved current reference

We suggest the following future works:

- Improve the accuracy of the reference voltage to make it more competitive to high-precision references.
- Reduce the chip area by improving the architecture and reducing the number of BJTs and resistors.
- The current reference part uses MOSFET as the core circuit, which is not very compatible with the BJT used in the voltage reference part, leading to a major deviation. A current reference with a BJT as the core should be developed to reduce the deviation.



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